Retail Payments

Easypaisa – Generate QR RSA API Integration Guide

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1. Easypaisa Payment Solution

Easypaisa is an electronic payment solution that easily and seamlessly integrates with any online website and/or shopping cart and enables users to pay online through Payment Gateway. Easypaisa is built upon the latest state of the art technologies and toolset that leverages many features a secure payment solution system must have. The users require no special technology or business license; only a valid email address to have an account on Easypaisa.

2. RSA Encryption

In order to use the web-services of OPS, it is necessary for the client to send a digital signature along with each API request. This signature is used by Easypaisa to authenticate the client before processing the request. The steps required to perform this encryption are described below:

Client and Easypaisa must exchange RSA keys before making API calls, and the length of RSA key must be 2048 bits. When making API call to Easypaisa, client uses the RSA private key to sign the API request. After receiving the API request, Easypaisa will use the client's RSA public key to verify whether the signature is matched to the content of API request. Similarly, when client receives the API response, it is highly recommended that client verifies the signature of API response by using Easypaisa's RSA public key.

2.1 Public/Private Key Exchange

An RSA key pair contains the private key and the public key. The private key is required for generating the signature, while the public key is used for verifying the signature. Many tools can be used to generate the RSA key pair. After the RSA2 key pair is generated, you must exchange the public key with the Easypaisa server for signature verification by completing the following steps:

1. **Upload your public key in Easypaisa merchant portal.**

Log in to the portal and visit the following path: "*Account Settings -> Public Key Configuration -> Upload Public Key*". Upload your public key here.

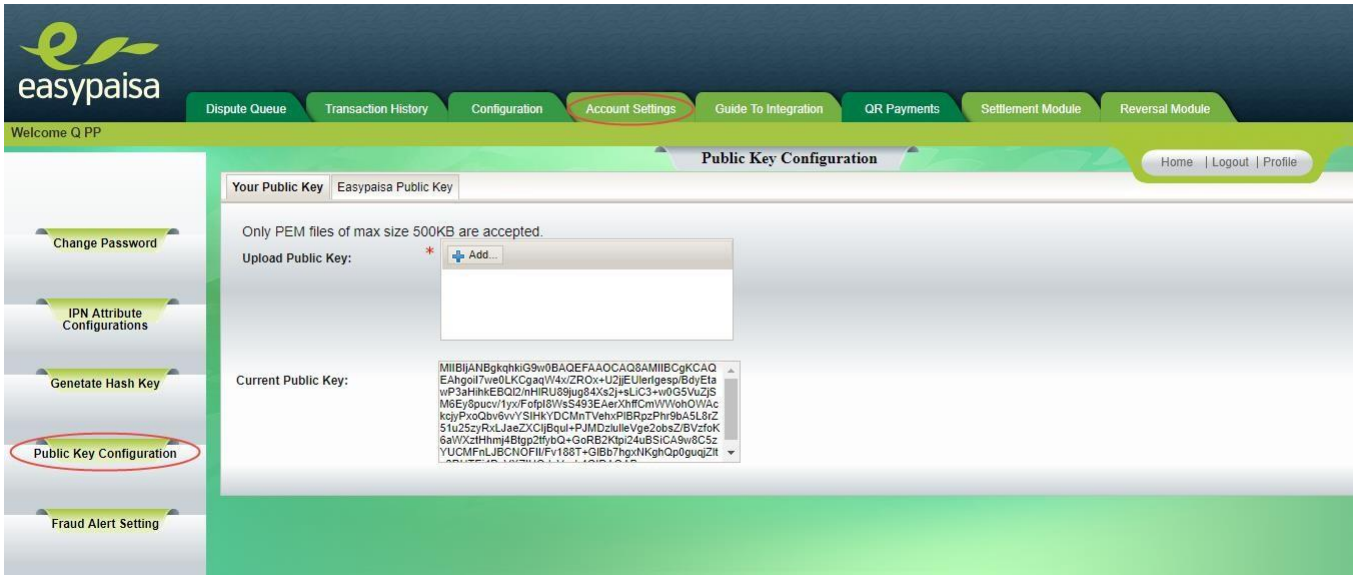

Figure 1 – Public Key Configuration (Upload)

2. **Download Easypaisa public key**. [OPTIONAL]

On the same path, enter the “Easypaisa Public Key” tab and click on the download button to download Easypaisa public key.

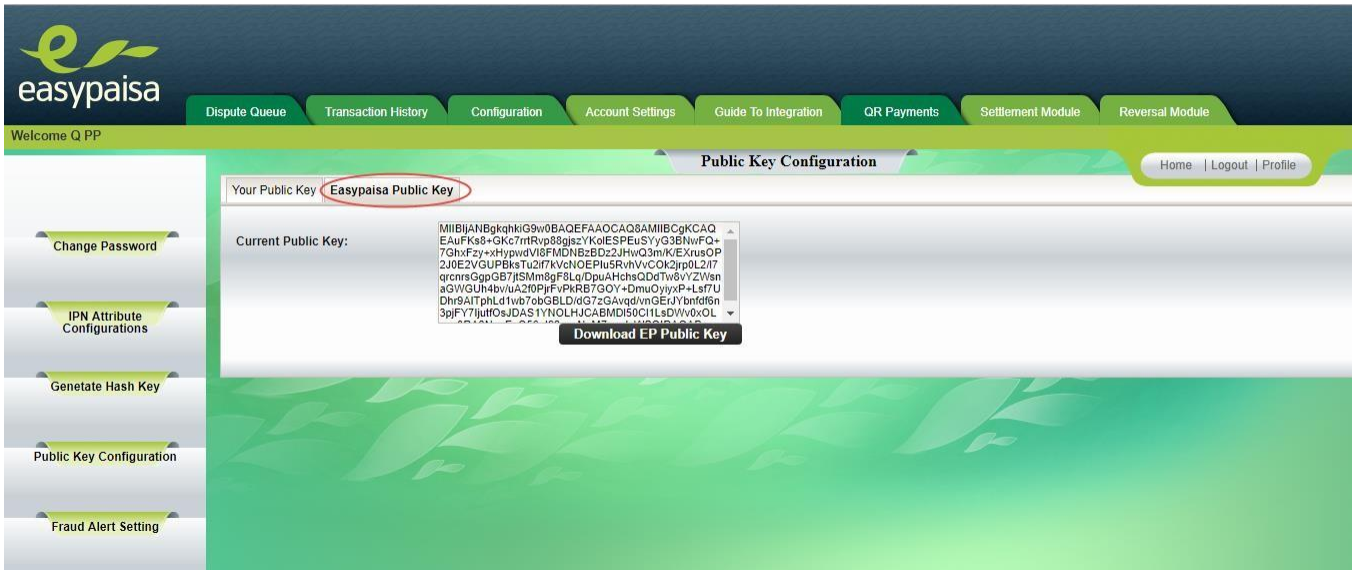

Figure 2 – Public Key Configuration (Download)

2.2 Signature Generation

The following figure illustrates a sample API request. The content enclosed by the two curly braces (inclusive) is what needs to be signed to create the RSA signature. And the generated signature is put in
the *signature* parameter.

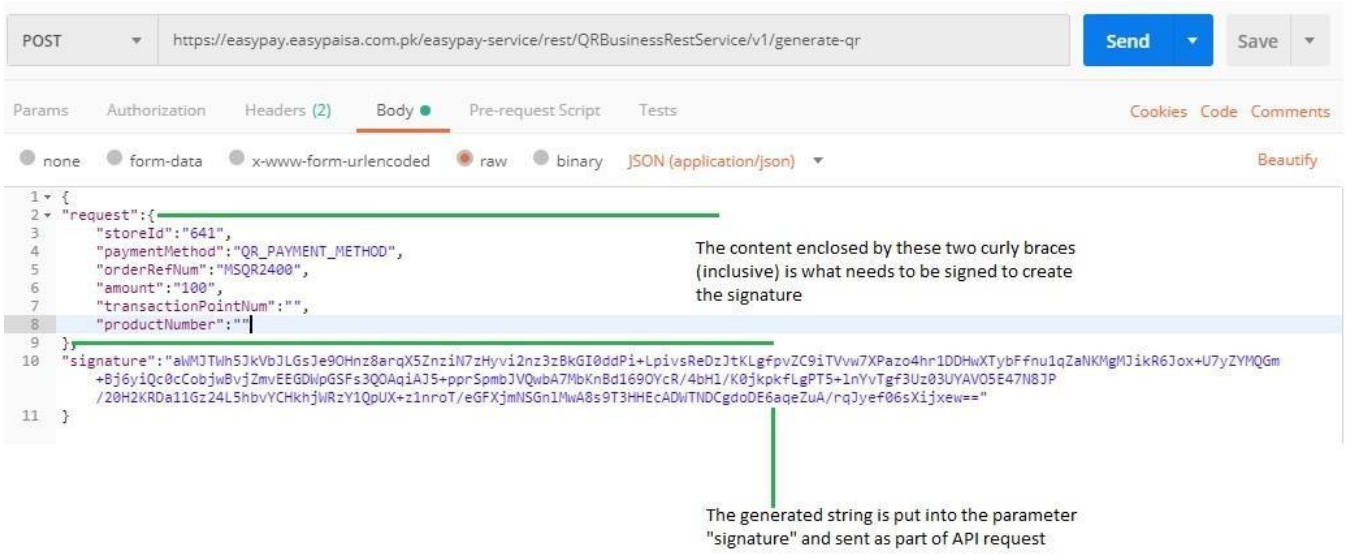

Figure 3 - Sample Request

1. Extract the content to be signed in JSON format and trim un-necessary spaces so that it looks like

   {"storeId":"641","paymentMethod":"QR_PAYMENT_METHOD","orderRefNum":"MSQR2400","amount":"100", "transactionPointNum":"","productNumber":""}

   this:

2. Hash the above mentioned JSON content by using the SHA256withRSA algorithm. Then use the client's RSA private key to sign the value to obtain the signature. To achieve a better security level, the length of the RSA key/pairs must be 2048 bits. The following sample


```
sXSGdA4tR4B0wnF6Nr8ShpfNMtHlSWuzLpdKoNWsklgCcspMVFFzgkez4BrmtYXkMqZh1tyz/8sNNU3F2VTuOfO
x+ZP4CnmVjNu8bciu4hTtnytA3Y1GZ/yXAUD+mVYGXeyDO3fHrqDdl7k3xV/Q8FCOdOculTsN6RRMxMZq+oqR2x
+EjOXVofp7BZJwhQE8Us4Al573FQz5EHDvCLx35lq2hoMhXLCVLVDBHcpCgotuY4mg91MXh/qqcFGY/hQKhGdWA
bZ9tcp2Q5s2ESh37rqxnOwZrrepK8NC6vMBPUiSPxQefpev2nGOJzAHOjRPbBXYw1XBS6RBYKAmufZZiQ==
```


   illustrates a raw signature:

   (This is sample)

3. Base64-encode the signature. A sample output is given below:


```
c1hTR2RBNHRSNElwd25GNk5yOFNocGZOTXRlSVNXdXpMcGRLbO5Xc2tJZONjc3BNVkZGemdrZXoOQnJtdFlYaO1
xWmgxdHl6LzhzTk5VMOYyVlR1MGZPeCtaUDRDbm1Wak51OGJjaXUOaFRObnlOQTNZMUdaL3lYQVVEK21WWUdYZX
lETzNmSHJxRGRJN2szeFYvUThGQO9kMGN1bFRzTjZSUk14TVpxK29xUjJ4KOVqT1hWb2ZwNOJaSndoUUU4VXMOQ
Ww1NzNGUXo1RUhEdkNMeDM1bHEyaG9NaFhMQ1ZMVkRCSGNwQ2dvdHVZNG1nOTFNWGgvcXFjRkdZL2hRS2hHZFdB
Ylo5dGNwMlE1czJFU2gzN3JxeG5Pd1pycmVwSzhOQzZ2TUJQVWlTUHhRZWZwZXYybkcwSnpBSE9qUlBiQlhZdzF
YQlM2UkJZSOFtdWZaWmlRPTO=
```


   (This is sample)

4. Use the obtained string as the value of the Signature parameter.

2.3 Signature Verification

A sample response is shown as below:

After receiving a response, perform the following steps to verify the signature:

1. Split the full response contents to 2 parts, the response JSON string and the signature string, by using the regular expression instead of JSON object.

2. Trim any un-necessary spaces (if any) from and hash the response JSON string by using the SHA256 algorithm to obtain a message digest. Be sure not to change the order of the response parameters.
3. Use the public key to decrypt the signature to a message digest.
4. Compare the two message digests obtained in step 2 and step 3. If the digests are the same, then it indicates that the signed data has not been changed.

3. Generate QR API

The Inquire Transaction web-service is a web-service that can be used to Inquire MA transactions in OPS. The parameters required by this API are mentioned below

3.1 Header

Header key	Header value	Description
Credentials	Base64Encoded username:password	Partner Account username & password for authentication

3.2 Request Parameters

Field Name	Description	Mandatory (M) / Optional (O)	Data Type
storeId	Store ID assigned by Easypaisa	M	String
paymentMethod	QR_PAYMENT_METHOD	M	String
orderRefNum	Merchant's system generated Order ID	M	String
amount	Total amount of the transaction in PKR. If this is left null, then a 'static QR' code will be generated.	O	String
transactionPointNum	The transaction point number assigned by Easypaisa	O	String
productNumber	An Identifier of the Product being purchased	O	String

3.3 Response Parameters

Field Name	Description	Data Type
qrCode	A stream of bits is returned. This bit stream is the QR Code that can be generated into an image	String
responseCode	A numeric code which will determine if the transaction is successful or state the reason for a failure	String
responseDesc	Possible values are : Code : Description SUCCESS : "0000" REQUIRED_FIELD_MISSING : "0002" INVALID_CREDENTIALS : "0010" INVALID_STORE_ID : "0006" SIGNATURE_ERROR : "0023" SIGNATURE_INVALID : "0024" INVALID_ORDER_ID : "0003"	Long

The QR Code bit stream that is returned by the API includes following information about the merchant and/or transaction:

- Payload Format Indicator (mandatory)
- Point of initiation method (optional)
- Merchant Identifier (mandatory)
- Merchant Category Code (mandatory)
- Transaction Currency Code (mandatory)
- Transaction Amount (optional)
- Country Code (mandatory)
- Merchant Name (mandatory)
- Merchant City (mandatory)
- Additional Data Field (mandatory)
    - Store ID (mandatory) – ID of that store
    - Order reference number (optional) – Order ID of the merchant as sent in the request.
    - Transaction Point ID (optional) – Necessary for retail merchants
    - Product Id (optional) – ID of the product as stored in EasyPaisa. For Partial QR codes.
- CRC (mandatory)

In order to convert the QR code into an image, developers can use any online plugin, freely

available on the web. Following is the URL to one possible plugin:

https://codebeautify.org/base64-to-image-converter

3.4 Web-services

Following are the URLs of Easypaisa QR Code APIs:

Environment	URL
Sandbox	https://Easypaisastg.easypaisa.com.pk/Easypaisa-service/rest/QRBusinessRestService/v1/generate-qr
Production	https://Easypaisa.easypaisa.com.pk/Easypaisa-service/rest/QRBusinessRestService/v1/generate-qr